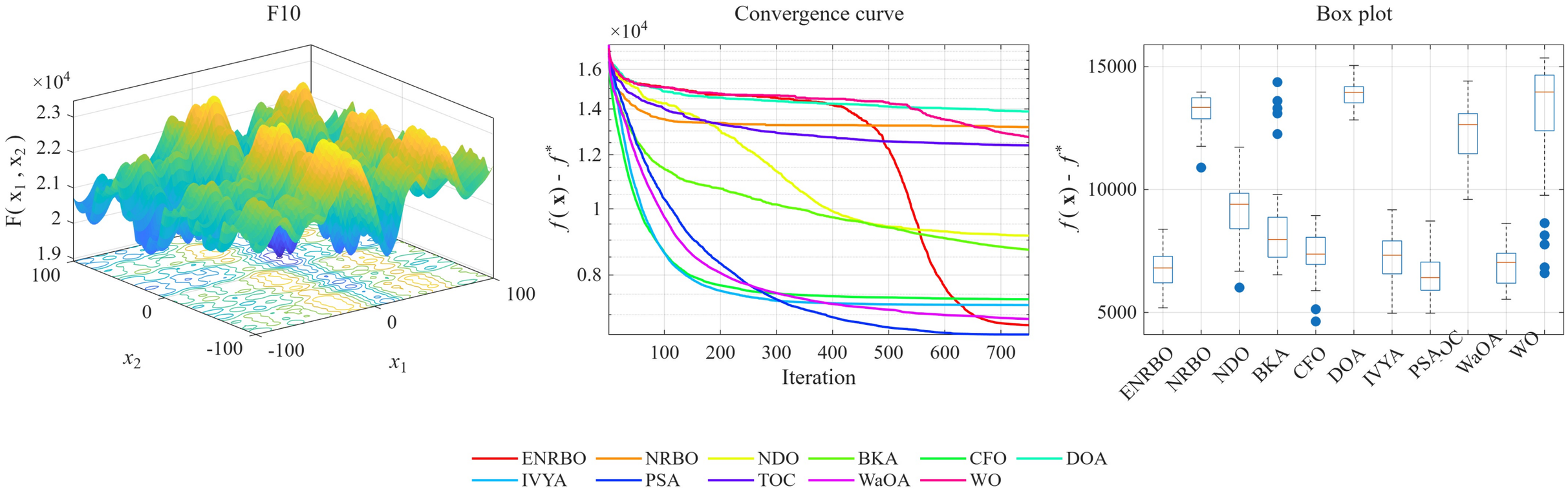

F10
×10^4
2.3
2.2
2.1
2
1.9
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^4
1.6
1.4
1.2
1
0.8
f (x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
15000
10000
5000
f (x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO